

Haider Dawood Khan

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Education

University of Michigan, Ann Arbor

Ann Arbor, Michigan

M.S. in Computer Science and Engineering

May 2028

- **Relevant coursework:** Reinforcement Learning, Human-AI Interaction

Lahore University of Management Sciences (LUMS)

Lahore, Pakistan

B.S. in Computer Science

May 2026

- **GPA: 3.97/4.0.** Graduated with **High Distinction**, ranked in the top 2% of the Computer Science Cohort.
- **Relevant coursework:** Deep Vision-Language Models, Advanced Topics in ML, Deep Learning, Computer Vision, Generative AI, Linear Algebra, Probability and Statistics, Calculus I & II.
- **Honors:** Joint 1st in the LUMS SBASSE batch (2022–24); 100% Merit Scholarship and Dean's Honor List (2023–26).

Publications

Noise In, Harm Out: A Harm-Weighted Eval of Noise Propagation in ASR-LLM Clinical Scribes

Ehsan Latif, *Haider Dawood Khan*, Shaheer Hammad, Dayyan Ali Akhtar, Agha Ali Raza (*Under Review, AAAI 2027*)

Research Experience

Research Assistant

May 2026 – Aug 2026

Lahore University of Management Sciences

- Developing a controller-guided latent-reasoning framework for VLMs that performs dynamic visual evidence insertion and recurrent latent reasoning, replacing explicit CoT before answer generation.
- Designed a multi-stage training pipeline comprising progressive distillation, cross-entropy guided beam search dataset construction, supervised controller training, and GRPO.
- Achieved **66.27% accuracy** on M3CoT test set with Qwen2-VL-2B-Instruct, outperforming explicit CoT by **14.67 percentage points** and the baseline latent-reasoning scheme by **3.4 percentage points**, while reducing average reasoning-trace length by **~5×**.

AI/ML Research and Development Intern

Jun 2025 – Sep 2025

Motive

- Used SAM-based instance segmentation to benchmark monocular depth models and metric-alignment transforms for vehicle distance estimation using radar distance supervision.
- Curated **200K+ metric-aligned image-dense depth map pairs** through a hybrid-binning pipeline with DINOv2 and DepthAnythingV2, refined via inter-model consensus.
- Reduced vehicle-distance MSE by **>20% relative to baseline** and extended YOLOX with a dense metric-depth prediction head.

Projects

SmolLM DPO-based Alignment via LoRA PEFT

- Built a lightweight decoder-only Transformer (RMSNorm, RoPE, GQA, SwiGLU) and integrated LoRA into attention and FFN modules, **training only ~1% of parameters** for alignment.
- Implemented DPO on Anthropic's HH-RLHF dataset with sample filtering, generation, and a pairwise preference dataloader; improved **preference accuracy by 16%** on held-out chosen-rejected pairs.

Model Compression: Pruning, Quantization & Distillation

- Performed unstructured pruning on VGG-11 to 70% sparsity, improving accuracy by **1.24%** with no MAC reduction; applied channel pruning to reduce MACs by **68.7%** and latency by **65%** while maintaining **91.84%** accuracy.
- Built PTQ and QAT pipelines across bitwidths for ResNet on CIFAR-100: FP16 and INT8 QAT were near-identical to FP32; MP matched baseline at **~4× smaller size**, with QAT outperforming PTQ.
- Evaluated distillation strategies with VGG-16/11 teacher-student pairs, comparing LM, DKD, Hints, and CRD: Hints achieved the best CIFAR-100 accuracy (**67.73%**), while LM minimized divergence to teacher logits.

Skills

Languages: Python, C/C++, SQL, JavaScript, TypeScript, Bash

Frameworks/Libraries: NumPy, Pandas, Matplotlib, PyTorch, TFLite, OpenCV, Hugging Face, Scikit-Learn, Streamlit, ChromaDB, Next.js, FastAPI, Flask, PostgreSQL

Cloud/Tools: AWS EC2/S3/Lambda, Codex, Git/GitHub, Linux, Jupyter, Postman, Firebase, Docker, Weights & Biases.